

Domain Knowledge-Guided Semi-Supervised Cluster-Then-Label Approach for Political Object Detection in the Wild

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Abstract

This paper introduces a semi-supervised framework for detecting politically relevant objects in real-world images, addressing the limitations of existing computer vision tools that fail to capture politically meaningful subcategories of generic objects. The approach integrates open-vocabulary object detection, image segmentation, unsupervised clustering, and domain-knowledge-guided labeling to reduce manual annotation costs while preserving accuracy and interpretability. The framework is demonstrated through two downstream applications using core statistical tools commonly employed in political science. First, a measurement exercise scales detected objects to produce *Objectscores*—image-based analogues to *Wordscores*—capturing the ideological valence of political symbols. Second, an associational analysis uses detected objects as variables in regression models of Black Lives Matter protest images to examine differences in media framing and the effects of visual features on perceived protest violence. The results show that conservative outlets disproportionately select images featuring police, and that police and fire heighten perceived violence, while defensive equipment such as shields attenuates it. Together, the applications demonstrate how automated political object detection can facilitate both ideological measurement and systematic analysis of visual political communication.

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1 Introduction

Visual political communication increasingly shapes public understanding of protests, campaigns, and political events. Yet, systematic analysis of visual data has lagged behind the textual domain because extracting politically meaningful features from images remains challenging. Off-the-shelf computer vision tools often fail to detect context-specific objects—such as police gear, protest signs, or fire—that carry political significance. As a result, researchers face a trade-off between costly manual annotation and coarse, automated measures that overlook key visual cues.

This paper introduces a semi-supervised framework for detecting politically relevant objects in real-world images, bridging this gap by integrating open-vocabulary object detection, image segmentation, unsupervised clustering, and domain-knowledge-guided labeling. The approach substantially reduces manual annotation costs while preserving accuracy and interpretability, offering political scientists a scalable tool for studying how visual information shapes political attitudes and behavior.

I demonstrate the framework’s utility through two distinct applications. First, as a measurement exercise, I apply ideological scaling to detected objects to derive Objectscores—image-based analogues to Wordscores—that quantify how liberal or conservative a political object is. This enables precise ideological measurement at the object level, linking visual features to well-established text-based scaling approaches. Second, as an associational analysis, I use the detected objects as covariates in regression models of Black Lives Matter (BLM) protest images to examine two questions: (1) how liberal and conservative media outlets visually frame protests, and (2) how specific visual features—such as police officers, fire, and protective gear—influence viewers’ perceptions of protest violence.

The results reveal systematic differences in visual framing across media outlets and highlight the perceptual consequences of politically charged objects. Conservative outlets are more likely to feature police officers, whose presence—along with fire—heightens perceived violence, whereas defensive gear such as shields mitigates it. Together, these applications demonstrate how automated political object detection can advance the study of visual political communication by enabling both

ideological scaling and causal inference in the analysis of protest imagery.

The paper proceeds as follows. Section 2 introduces the semi-supervised political object detection framework, detailing how open-vocabulary detection, segmentation, clustering, and domain-knowledge-guided labeling jointly minimize annotation costs while maintaining accuracy. Section 3 presents the first application, where the detected political objects are used to construct *Objectscores*—image-based analogues to *Wordscores*—to quantify the ideological valence of visual elements in protest images. Section 4 turns to the second application, using the detected objects as independent variables in regression analyses to examine media portrayals of Black Lives Matter protests and their effects on perceived protest violence. Section 5 discusses the broader methodological and substantive implications of this approach, and Section 6 concludes.

2 Method

2.1 Related work

Research on object detection has evolved rapidly over the past decade, moving from rigid, closed-set detection toward more flexible, language-guided approaches. Early Closed Set Object Detection (CSOD) models were designed for fixed, pre-defined categories (e.g., “person,” “hat”) and achieved impressive performance on benchmarks like COCO and PASCAL VOC. However, these models were inherently limited: any politically meaningful subcategories—such as distinguishing a “MAGA hat” from a generic red hat—were absent from training sets. Incorporating such categories would require costly, large-scale manual annotation, making CSOD approaches ill-suited for political science research, where rare, domain-specific objects often carry critical ideological meaning.

To address these constraints, Open Set Object Detection (OSOD) methods emerged, allowing models to recognize unseen categories beyond the training set. Yet, OSOD models typically require fine-tuning for each new class, risking catastrophic forgetting—the loss of previously learned categories. This limitation is particularly problematic for political applications, where researchers may

wish to add new politically relevant subcategories (e.g., police officers, police helmets) without sacrificing performance on general objects like “person” or “helmet.”

More recently, Open Vocabulary Object Detection (OVOD) and Referring Expression Comprehension (REC) approaches have leveraged Vision-Language Models (VLMs) to enable zero-shot detection guided by natural language prompts. OVOD detects all instances of a given category, while REC focuses on locating a single object described in detail (e.g., “the rightmost person wearing a yellow t-shirt”). Despite their flexibility, both approaches rely heavily on linguistic descriptions. As a result, they struggle with rare or ideologically nuanced objects that are difficult to capture in text (Jiang et al. 2024) — precisely the kinds of objects central to political science research. For example, as shown in Figure 1, even the state-of-the-art Grounding DINO v2 misidentifies or misses many “Make America Great Again” hats—detecting only two frontal hats correctly while missing non-frontal ones, mislabeling others, and producing incomplete text outputs (e.g., labeling MAGA hats as “america great hat”). This highlights the limitations of current OVOD and REC tools for detecting politically relevant objects.



Figure 1: OVOD and REC result from GroundingDINOv2 when asked to detect “Make America Great Again hat” in an image from January 6 U.S. Capitol Attack

2.2 Framework: Domain knowledge-guided cluster-then-label approach

To overcome this lack of flexibility in detecting politically meaningful objects, I propose a domain knowledge-guided, cluster-then-label framework. This semi-supervised approach enables detection of rare political objects beyond standard categories, while also reducing annotation costs and preserving high accuracy on real-world images. The framework proceeds through seven stages, detailed below and visualized in Figure 2.

In-distribution image collection: In-distribution image collection refers to gathering data that comes from the same distribution as the deployment data. Political scientists are typically interested in in-the-wild images used in political communication such as news images, political campaign materials, and social media posts. Therefore, collecting in-the-wild images for training can improve model performance in the wild. Collection of in-the-wild images featuring political objects can be done by identifying real-world events with higher probabilities of seeing the objects of interest. For example, collecting in-the-wild images of MAGA hats can be done by collecting images from the January 6 U.S. Capitol Attack in which strong Trump supporters convened. Collecting in-the-wild images of Pussy hats and police headgear can be done by collecting images from the 2017 Women’s March, and police protester confrontations, respectively.

OVOD: The second stage is to detect all instances of an umbrella object using OVOD. For example, if interested in detecting MAGA hats, you ask the OVOD model of your choice to detect any hat in provided images. Given a set of images and objects of interest, the model outputs bounding boxes for each instance of each object.

Image Segmentation: IS is the process of partitioning an image into multiple segments or regions to identify and isolate meaningful parts. In this stage, IS is used to filter out the background for each bounding box from the second OVOD stage. Image segmentation is essential for in-the-wild political image data, as backgrounds such as humans wearing hats are likely to be highly correlated with objects. For example, MAGA hats are highly likely to be worn by white males, and Pussy hats are highly likely to be worn by females. A model trained on unsegmented images may learn patterns in the background and use that information to make predictions.

Clustering: The fourth stage is clustering based on the reduced Deep Learning (DL) features of the segments from the third stage. UMAP is recommended for dimensionality reduction of deep learning image features as it offers several advantages compared to older techniques like PCA and t-SNE, including a good balance between preserving data structure and computational performance. As for clustering, the HDBSCAN algorithm is preferred for this case. Unlike K-means or GMM, HDBSCAN does not require a researcher to specify the number of clusters in advance. Unlike DBSCAN, HDBSCAN can detect clusters with varying densities and does not require a distance threshold, which is not intuitive and hard to tune.

Domain knowledge-guided labeling: The fifth stage utilizes domain knowledge and queries used in the first data collection stage in evaluating the quality of clusters from the fourth clustering stage. Oftentimes, clustering results do not align with human expectations. Rather than taking the clustering results at the face value, I propose evaluating the quality of clusters based on domain knowledge and adjusting and labeling the clusters accordingly. The evaluation is case-specific, and thus further discussed under the Experiment section.

Training & prediction: The last two stages are training and prediction. In this stage, labels created in the fifth stage by combining domain knowledge and clustering results are used for training.

2.3 Experiment

In this experiment, I focus on detecting politically meaningful head wear, specifically MAGA hats, Pussy hats, and police headgear.

Data: In the first stage of in-distribution image collection, three queries are used to collect real-world images featuring the three objects: “January 6 U.S. Capitol Attack”, “2017 Women’s March”, and “police protester confrontation.” Then, SOTA OVOD and IS models - Grounding DINOv2 and SAM2 - are used to detect and segment all instances of headwear. Table 9 in Appendix shows the number of collected images and detected head wear segments for each query. Segment counts are greater than image counts because an image can contain multiple instances of head wear.

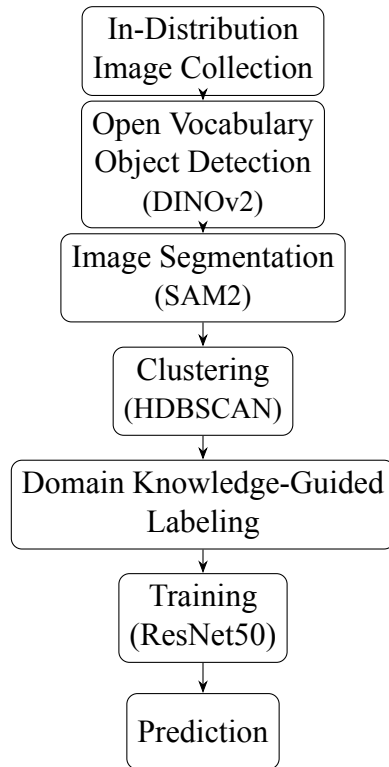


Figure 2: The framework of domain knowledge-guided cluster-then-label approach for political object detection in the wild

Domain Knowledge: Figure 3 shows a simplistic view of domain knowledge related to this specific experiment. Political scientists know that January 6 Capitol Attack was a violent protest with conservative participants and had a higher probability of police-protester confrontation. They also know that the 2017 Women’s March was a peaceful protest with liberal participants and had a lower probability of police-protester confrontation. Due to these differences, it is hard to see the same politically relevant head wears in the two events. To be more specific, it is hard to see MAGA hats in the Women’s March, and it is hard to see Pussy hats in the Capitol Attack. This domain knowledge is used to evaluate the quality of clusters produced by the algorithm.

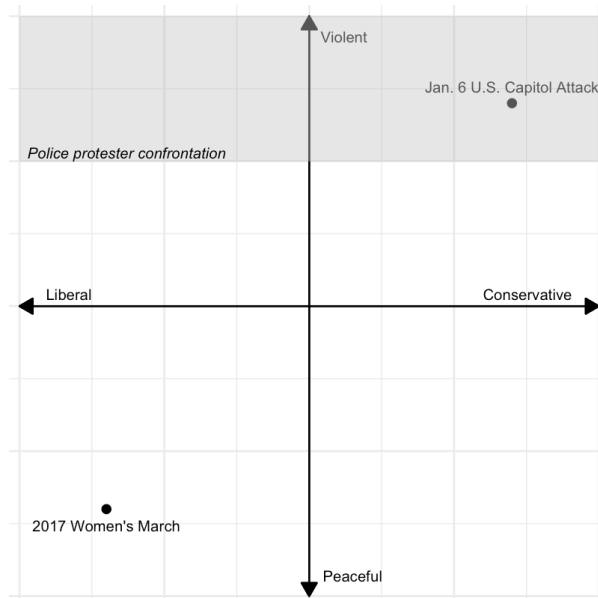


Figure 3: A simplistic view of domain knowledge. The horizontal dimension indicates the protester ideology, and the vertical dimension indicates the level of protest violence.

Domain Knowledge-Driven Cluster-then-Labeling: Figure 4 shows HDBSCAN-clustering results on UMAP-reduced ResNet50 embeddings. Part (a) shows that Cluster 0 mostly consists of segments from Women’s March and Capitol Attack, and Cluster 1 mostly consists of segments from Capitol Attack and police confrontation. Cluster 2 consists of segments from all three queries. Part (b) shows that Cluster 0 (the second row) contains both MAGA and Pussy hats due to their similarity in color. Based on the domain knowledge, we know this cluster needs to be divided into two: MAGA and Pussy. Cluster 1 in the third row mostly consists of police helmets and thus does

not require adjustment. Cluster 2 consists of hats without any political meanings and thus should not be treated a valid cluster.

Query	-1	0	1	2
Women’s March	106	811	2	652
Capitol Attack	359	666	187	1047
police confrontation	206	34	193	662

(a) Cluster distribution by query.



(b) Samples from clusters.

Figure 4: HDBSCAN-clustering results. (a) Distribution of samples across clusters for each query. Valid clusters are 0 through 2. -1 is noise. (a) Samples from clusters. The rows shows samples from -1, 0, 1, and 2, in that order.

Based on domain knowledge-guided evaluation of clusters, one can develop a labeling Algorithm 1 in Appendix. Noise and Cluster 2 segments are labeled as “Unknown”. Cluster 1 segments are labeled as “Police.” Cluster 0 is divided and labeled based on originating queries. If originated from Women’s March, labeled as “Pussy”; if from Capitol Attack, labeled as “MAGA.” The process is also visualized in Figure 8 in Appendix.

2.4 Evaluation

The proposed labeling approach is compared against two alternative labeling approaches. The first alternative labeling approach is to label as queries. This approach labels all head wears detected in images as the originating queries. For example, all head wears from Women’s March query are labeled as Pussy hats; all head wears from Capitol Attack query are labeled as MAGA hats; all head wears from the images collected with the police-protester confrontation query are labeled as “police.” The second alternative labeling approach is to label as clustering results. This approach blindly accept clustering results from the algorithm as meaningful and label segments accordingly, as shown in Table 1.

To isolate the effect of labeling approach, I develop three models using the same embeddings and dimensionality reduction and clustering algorithms. The only element that differs is the labeling

Model	Training data	Label by
Proposed	Real-world	Cluster assignments + domain knowledge
Alternative 1	Real-world	Query
Alternative 2	Real-world	Cluster assignments

Table 1: Comparison of models.

approach. I test the models on the same test data of about 1,000 segments from in-the-wild protest images.

The models are evaluated both qualitatively and quantitatively. For qualitative evaluation, the distribution of predictions by query and sample images are provided for soft-check the match with domain knowledge. For quantitative evaluation, two measures - an average within-cluster variance score and a modified F1 score - are provided for each model. Average within-cluster variance is calculated by computing the mean squared distance between each point and its cluster centroid, then averaging these variances across all clusters. The smaller the average within-cluster variance, the similar the visual features of segments within clusters.

The modified F1 score is different from the standard one in that it uses only a subset of predictions. The standard F1 score is a function of precision and recall computed with all predictions. Computing precision and recall with all predictions is impossible without complete ground truth labels.

I handle the lack of ground truth labels by using weak, domain knowledge-driven labels. These labels are weak for two reasons. First, they come from external metadata (queries) rather than the internal visual features from segments. Second, labels are incomplete. Labels exist for only a subset of predictions with high probabilities of being true or false based on domain knowledge.

For example, based on the domain knowledge in Figure 3, we are sure that no segments from Women’s March should receive a MAGA hat prediction and that no segments from Capitol Attack should receive a Pussy hat prediction. We also sure that few segments from the 2017 Women’s March should receive police headgear predictions, as it was a peaceful protest. However, we do not have such confidence for Capitol Attack, as it involved participants’ violent actions and resultant police involvement. Therefore, I do not use these predictions to construct the F1 score. I only use

predictions with high probabilities of being true or false based on the domain knowledge as shown in Equation 1.

$$\begin{aligned}
F1 &= \frac{1}{O} \sum_{o=1}^O F1_o \\
F1_o &= \frac{2 \times \text{Precision}_o \times \text{Recall}_o}{\text{Precision}_o + \text{Recall}_o} \\
\text{Precision}_{MAGA} &= \frac{\text{MAGA}(\text{Jan 6})}{\text{MAGA}(\text{Jan 6}) + \text{MAGA}(\text{Women})} \\
\text{Recall}_{MAGA} &= \frac{\text{MAGA}(\text{Jan 6})}{\text{MAGA}(\text{Jan 6}) + \text{Pussy}(\text{Jan 6})} \\
\text{Precision}_{Pussy} &= \frac{\text{Pussy}(\text{Women})}{\text{Pussy}(\text{Women}) + \text{Pussy}(\text{Jan 6}) + \text{Pussy}(\text{Police})} \\
\text{Recall}_{Pussy} &= \frac{\text{Pussy}(\text{Women})}{\text{Pussy}(\text{Women}) + \text{MAGA}(\text{Women})} \\
\text{Precision}_{Police} &= \frac{\text{Police}(\text{Police})}{\text{Police}(\text{Police}) + \text{Police}(\text{Women})} \\
\text{Recall}_{Police} &= \frac{\text{Police}(\text{Police})}{\text{Police}(\text{Police}) + \text{Pussy}(\text{Police})} \tag{1}
\end{aligned}$$

The modified F1 score is the average F1 scores of individual objects. O is the number of objects of interest. In this experiment, O is 3 as we are interested in detecting three head wears: MAGA hats, Pussy hats, and police headgear. I take Precision_{MAGA} and Recall_{MAGA} for MAGA hat to explain how precision and recall are constructed with only a subset of labels. I treat MAGA hat predictions only from Capitol Attack images as true positives, as they have a high probability of being so. In comparison, MAGA hat predictions from police-protester confrontation images do not have such a high probability. Similarly, I treat MAGA hat predictions only from Women’s March as false positives, as they have a high probability of being so. I also treat Pussy hat predictions only from Capitol Attack as false negatives, as they have a high probability of being so. Essentially, the modified F1 score measures how close the predictions are to the domain knowledge.

2.5 Results

2.5.1 Qualitative

Proposed: Figure 5 shows (a) the distribution of class predictions by query and (b) the prediction samples. From (a), we can see that most MAGA hats and Pussy hat predictions are from Capitol Attack and Women’s March, respectively, which aligns with our domain knowledge. We also see that few police headgear predictions are from Women’s March and quite many from Capitol Attack, as expected. From (b), we see three politically meaningful clusters of head wear. The second row shows red MAGA hats, the third police headgears, and the fourth pink Pussy hats. Both the distribution of predictions and sample images indicate the model trained with the proposed framework has great performance in the wild. Such result is a great improvement from alternatives that follows.

Query	Unknown	MAGA	Police	Pussy
W March	139	5	2	153
January 6	293	124	42	4
Police	185	2	35	1

(a) Prediction distribution by query



(b) Prediction samples

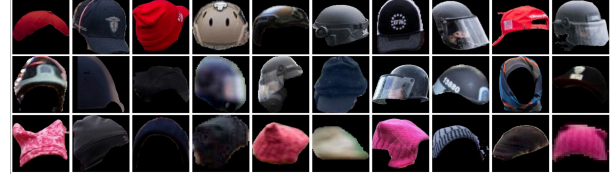
Figure 5: Classification results from the proposed model. (a) Distribution of predictions for each query. (b) Visual samples from predictions. The rows shows samples from "Unknown", "MAGA hat", "Police", and "Pussy hat" predictions, in that order.

Alternative 1: Query labels: Figure 6 shows results from the first alternative labeling approach: labeling as queries. From (a), we can see most MAGA hat, Pussy hats, and police head wear predictions are from Capitol Attack, Women’s March, and police-protester confrontations, respectively, although not perfect. We still see 41 MAGA hat predictions are from Women’s March, and 38 predictions are from Capitol Attack. Although the distributions shown in (a) match our expectation, predictions samples in (b) do not. The first, second, and third rows in (b) show MAGA hat, police head wear, and Pussy hat predictions, respectively. However, we see many hats that

do not belong to any of the three categories exist in all three categories of predictions. This is unsurprising because this alternative labeling approach - labeling all detected head wears by query - did not effectively handle politically irrelevant head wears in real-world images by either labeling them as “Unknown” or excluding them in the training set.

Query	MAGA	Police	Pussy
Women’s March	41	29	244
January 6	374	40	38
Police	27	155	37

(a) Prediction distribution by query



(b) Prediction samples

Figure 6: Classification results from the first alternative model. (a) Distribution of predictions for each query. (a) Visual samples from predictions. The rows shows samples from ”MAGA hat”, ”Police”, and ”Pussy hat” predictions, in that order.

Alternative 2: Cluster labels Figure 7 shows results from the second alternative labeling approach: labeling as clustering results. From (a), we see undesirable distributions of predictions. Cluster 0 predictions are nearly equally from two politically distinct events, Women’s March and Capitol Attack. Cluster 2 predictions are from all three queries. Part (b) shows that Cluster 0 shown in the second row consists of both MAGA and Pussy hats. Cluster 2 consists of black hats without any political meanings.

Query	-1	0	1	2
Women’s March	26	162	0	111
January 6	65	133	43	222
Police	54	3	34	132

(a) Prediction distribution by query.



(b) Prediction samples.

Figure 7: Classification results from the second alternative results. (a) Distribution of predictions across clusters for each query. Valid clusters are 0 through 2. -1 is noise. (b) Visual samples from predictions.

2.5.2 Quantitative

Table 2 shows that the model trained with the proposed domain knowledge-guided cluster-then-labeling approach has a lower average within-cluster variance score and a higher F1 score than the alternative “labeling as queries” approach. That means the predictions from the proposed model have greater visual similarities within clusters and a closer match to the domain knowledge. The second alternative model has a slightly smaller within-cluster variance score than the proposed model because the model is designed to minimize within-cluster variance without considering the match with domain knowledge. This model does not have an F1 score because the clusters produced from this model are not politically meaningful and thus cannot be labeled as politically relevant objects.

Model	Within-cluster variance	F1
Proposed	1.90	0.968
Alternative 1: Query labels	3.32	0.882
Alternative 2: Cluster labels	1.69	N/A

Table 2: Quantitative results for three labeling approaches.

3 Application 1: *Objectscores*

3.1 Introduction

Political scientists have been interested in placing individuals and organizations along the liberal-conservative political ideology. Poole and Rosenthal (2000) developed DW-NOMINATE scores to indicate how conservative or liberal a congressperson is based on their roll-call votes ; Bonica (2014) developed CF scores to indicate how conservative or liberal Political Action Committees (PACs), interest groups, and small donors are based on how much they donated to whom. Their interest goes beyond individuals and organizations and expands to texts and words. Benoit and Laver (2003) developed *Wordscores* to indicate which documents or words are more conservative or liberal. These scores are used by numerous other scholars. DW-NOMINATE scores have been

used to measure elite polarization and its subsequent effects on mass polarization and income inequality (McCarty, Poole, and Rosenthal 2016); CF scores to analyze how campaign contributions affect legislative behaviors (Barber 2022); Wordscores to estimate party positions based on their manifestos, especially in multi-party countries, like Germany (Linhart and Shikano 2009).

Despite the similarity between text and images as high-dimensional data carrying latent meanings, analogous tools for images remain underdeveloped. For text, tokenizers, stemmers, and lemmatizers—tools that break text into meaningful units—have existed since the 1950s, with extensive discussions on how these preprocessing choices shape analysis (Denny and Spirling 2018). For images, however, comparable tools to segment visual data into theoretically meaningful units are largely absent.

This paper proposes one such tool: a visual tokenizer for politics. Specifically, I develop a political object detection framework that breaks images into politically meaningful objects rather than pixels or generic object categories. Existing approaches either focus on low-level pixels, as in Convolutional Neural Networks (CNNs), or on semantically meaningful objects, as in standard object detection models. Both have limitations. Pixel-based methods lack interpretability, as it is difficult to ascribe theoretical meaning to sets of pixels flagged as important. Off-the-shelf object detectors identify semantically meaningful objects, like hats or flags, but cannot distinguish politically salient subcategories such as MAGA hats, BLM flags, or police officers.

Breaking images into semantically and politically meaningful objects offers two advantages. First, it improves interpretability: rather than guessing what a cluster of pixels represents, researchers can directly interpret objects with clear political relevance. Second, it enables theory testing: distinguishing between police officers and civilian protesters, or between MAGA and non-MAGA hats, allows scholars to evaluate hypotheses about media bias or the visual portrayal of state power in protests. Without such distinctions, important questions—for instance, whether conservative media depict protests as more violent by emphasizing police presence—cannot be rigorously tested.

Building on this framework, I introduce *Objectscores*, the image-based analogue to *Wordscores*

developed by (Benoit and Laver 2003). The logic parallels that of text: in Wordscores, each reference text is assigned a known ideological position (e.g., left or right). A word’s score is then calculated as the weighted average of these positions, with weights given by word frequencies in the reference texts. Words appearing primarily in left-leaning texts receive left-leaning scores; those in right-leaning texts receive right-leaning scores; and words common to both fall in between. In *Objectscores*, I replace texts with images and words with politically meaningful objects detected using my domain knowledge-guided, semi-supervised cluster-then-label approach. This enables the same statistical machinery used in textual analysis to quantify the ideology of visual objects.

3.2 Analysis

Objectscores can be computed in two different ways, each with distinct advantages and disadvantages. The first approach is the log-odds score, which measures the relative frequency of an object in Conservative versus Liberal reference texts. This is calculated by taking the log of the ratio of the conditional probabilities $P(t|Conservative)$ and $P(t|Liberal)$ as shown in Equation 2. If an object appears much more often in Conservative images, its score will be strongly positive; if it is more common in Liberal images, the score will be strongly negative. This measure highlights the distinctiveness of objects because it captures how disproportionately associated a term is with one side compared to the other. However, it can be unstable for rare objects, since very low probabilities can cause the ratio to blow up to extreme values, and the log-odds scale is less intuitive for non-technical audiences.

$$LogOddsScore(o) = \log\left(\frac{P(o|Conservative)}{P(o|Liberal)}\right) \quad (2)$$

The second approach is the weighted probability score, which averages the probabilities of an object across the two reference sets using their ideological labels (+1 for Conservative, −1 for Liberal) as weights, as shown in Equation 3. In this formulation, an object exclusively used by Conservatives will receive a score near +1, while one used only by Liberals will be near −1. If an

object appears frequently in both camps, its score falls closer to the center, reflecting a balanced association. This method is easier to interpret because the scores are bounded and directly correspond to ideological placement, making it more accessible when communicating results. The drawback is that it does not emphasize distinctiveness as strongly as the log-odds method. Objects that are widely used by both camps may still receive a score leaning liberal or conservative, even if they are not meaningful for differentiating between the two. Only when usage is nearly equal do such objects fall in the center of the scale.

$$\text{WeightedProbabilityScore}(o) = P(o|\text{Conservative}) * 1 + P(o|\text{Liberal}) * -1 \quad (3)$$

In short, the log-odds score excels at identifying distinctive ideological markers by highlighting relative frequency differences, while the weighted probability score provides a more interpretable and intuitive measure of ideological direction. The choice between them depends on whether the goal is to uncover sharp ideological signals or to present results in a form that is easily understandable and comparable across objects.

3.3 Data

I construct *Objectscores* using domestic protest images collected from Bing Images. Search keywords were derived from the largest U.S. protests since 2017, as identified by Ulfelder (2025). Table 3 lists all keywords used in the image search. In total, about 5,000 protest images were collected: 2,664 from liberal protests and 2,116 from conservative protests. Images from liberal protests were assigned a label of -1 , and images from conservative protests were assigned a label of 1 .

Using the proposed domain knowledge-guided cluster-then-label approach, I developed a customized political object detection tool designed to identify politically meaningful subcategories of four general object classes—headwear, flags, signs, and buildings—as well as other general ob-

Liberal (Labeled as -1)	Conservative (Labeled as 1)
March for Life	Proud Boys
Pro-Choice	Pro-Trump
Women Lives Matter	Back the Blue
Pro-Palestine	Pro-Life
Women’s March	Confederacy
Anti-Trump	MAGA
Pride	Anti-Vax
Black Lives Matter	White Lives Matter
Planned Parenthood	1776 Restorative Movement
Drag Queen Story Hour	
Gaza	

Table 3: Keywords for Image Search

jects that do not require further subcategorization. Table 4 summarizes the subcategories defined for each object class.

From the detected political objects, I construct an *image-object matrix*, the visual analogue of a *document-term matrix* in text analysis. In this matrix, rows correspond to images and columns to detected objects. The resulting political object matrix includes all 5,000 images (rows) and 31 political object categories (columns). The list of 31 political object categories can be found in Table 4.

3.4 Results

Tables 5 and 6 present the most liberal and conservative objects identified using two alternative *Objectscores*: log-odds and weighted probability scores. In both measures, smaller values indicate stronger associations with liberal protests, while larger values indicate stronger associations with conservative protests.

According to the log-odds results in Table 5, the most liberal objects include Pussy hats, rainbow flags, and Trans Rights Now signs, whereas the most conservative objects include orange beanies, Confederate flags, and national American flags. These findings align well with expectations: Pussy hats, rainbow flags, and Trans Rights Now signs have been highly visible in feminist and LGBTQ rights demonstrations, while orange beanies are associated with the far-right Proud Boys, and Con-

7 General Objects (w/o Subcat.)	Hat / Helmet (w/ 15 Subcat.)	Flag (w/ 5 Subcat.)	Signs / Banners (w/ 8 Subcat.)	Buildings (w/ 2 Subcat.)
man woman boy girl beard drag fire	Orange beanie Bright green hat Police hat Blue hat Beret Proud Boys hat Black beanie March for Life beanie Ski mask White bike helmet MAGA hat American beanie Trump hat Pussy hat Unknown	Rainbow American March for Lives Confederacy Unknown	Anti-abortion [Socialist Alternative Fight Racism] Box Pro-Trump Traffic sign [Don't Let McConnell Trump] Trans Rights Now Unknown	White House Unknown

Table 4: List of objects that can be detected by my customized political detection tool. Objects in square brackets are divided into two lines to fit the table.

federate and American flags are common symbols in conservative protests, including the January 6 insurrection.

One less intuitive result is the appearance of police hats among the most liberal objects. Mechanically, this reflects that police hats occur more frequently in liberal than in conservative protest images. However, this should not be interpreted as evidence of greater police presence at liberal protests. Police attire varies across contexts: officers often wear hats in routine or less confrontational situations, helmets when anticipating violence, or bike helmets when regular helmets are scarce during large-scale events. If this pattern holds, the association of police hats with liberal protests—contrasted with helmets and bike helmets linked to conservative ones—may suggest that liberal demonstrations generally involved less violent interactions with police, whereas conservative protests prompted preparations for more confrontational encounters. Exploring this hypothesis further, however, falls beyond the scope of this paper.

The weighted probability results in Table 6 largely overlap with the log-odds findings: Pussy hats, rainbow flags, American flags, Confederate flags, and orange beanies remain strongly associ-

Liberal Object	Log Odds	Conservative Object	Log Odds
pussy hat	-3.343	orange beanie	3.445
rainbow flag	-2.857	confederate flag	1.712
trans rights now sign	-2.363	american flag	1.400
march for life beanie	-1.870	helmet	1.385
police hat	-1.407	ski mask	1.265
bright green hat	-1.207	white bike helmet	1.115
pro abortion sign	-0.882	maga hat	1.024
march for lives flag	-0.635	proud boy hat	0.696
black beanie	-0.551	unknown flag	0.425
blue hat	-0.465	american beanie	0.422

Table 5: Top Liberal and Conservative Objects with Log-Odds Scores (rounded to three decimals)

ated with the same ideological camps. However, some objects absent from the log-odds rankings—such as unknown signs and unknown flags—also emerge here. This difference arises because the weighted probability metric emphasizes absolute frequency differences between liberal and conservative images, whereas the log-odds metric highlights objects that are most distinctive to each side, regardless of how frequently they appear overall.

Liberal Object	P_{lib}	P_{con}	Score	Conservative Object	P_{lib}	P_{con}	Score
unknown sign	0.247	0.176	-0.071	unknown flag	0.089	0.136	0.047
pussy hat	0.013	0.000	-0.012	american flag	0.011	0.045	0.034
rainbow flag	0.010	0.001	-0.010	helmet	0.003	0.012	0.009
capitol hill building	0.105	0.096	-0.009	maga hat	0.004	0.010	0.007
pro abortion sign	0.008	0.003	-0.005	proud boy hat	0.005	0.011	0.005
police hat	0.004	0.001	-0.003	confederacy flag	0.001	0.006	0.005
bright green hat	0.003	0.001	-0.002	unknown headwear	0.452	0.457	0.004
blue hat	0.005	0.003	-0.002	orange beanie	0.001	0.003	0.003
unknown building	0.010	0.009	-0.001	ski mask	0.001	0.003	0.002
march for lives flag	0.002	0.001	-0.001	cowboy hat	0.004	0.005	0.001

Table 6: Top Liberal and Conservative Objects with P_{liberal} , $P_{\text{conservative}}$, and Weighted Probability Scores (rounded to three decimals)

Taken together, the log-odds and weighted probability results complement each other. The log-odds approach identifies objects that are distinctive to liberal or conservative protests regardless of overall frequency, which explains why symbols like the Trans Rights Now sign or orange beanies rank highly even if they appear relatively infrequently. In contrast, the weighted probability mea-

sure elevates objects that show large absolute frequency differences between the two ideological camps, even if those objects occur across both. This explains why items such as unknown signs and unknown flags emerge only in the weighted probability rankings. Using both measures provides a fuller picture: log-odds highlights symbolic distinctiveness, while weighted probabilities capture prevalence-driven contrasts.

3.5 Usages

The availability of *Objectscores* opens up a wide range of substantive questions at the intersection of symbols, political communication, and political behaviors. First, *Objectscores* allow scholars of symbols to identify which objects are tied to ideological identities in a systematic and scalable way. *Objectscores* can be also used to study ideological drift and contestation over time. Political objects often change in meaning depending on their context (e.g., the American flag, face masks), and longitudinal analyses of ideology scores can reveal how particular symbols are appropriated, contested, or re-signified during key political moments.

Second, scholars of framing and political communication can use *Objectscores* to investigate how the ideological composition of visual content varies across media sources, platforms, and geographic contexts. For instance, one could compare whether conservative and liberal news outlets feature different sets of political objects when covering the same protest event, or whether the prevalence of certain symbols differs systematically between urban and rural protests. This enables researchers to quantify ideological framing in visual content with far greater precision than was previously possible.

Third, object-level ideology scores facilitate experimental research on visual persuasion. A scalable, object-level scoring system allows researchers to design experiments that manipulate specific ideological cues embedded in images while holding everything else constant. This improves both ecological validity—because real-world symbols are used—and treatment control, since objects can be isolated and varied systematically. Such experiments can test whether certain symbols trigger stronger partisan reactions, influence perceptions of protest legitimacy, or shift opinions on

related policy issues.

Finally, *Objectscores* open the door to multimodal research linking visual and textual political communication. For example, scholars could study whether images and accompanying text in news articles convey consistent ideological signals—or whether visual content diverges from textual framing in systematic ways. This can help address broader questions about how visual and verbal political rhetoric jointly shape public opinion.

4 Application 2: Objects as variables in regression

4.1 Introduction

Building on the previous application, this section shifts from *scoring* individual objects to using them as variables in regression analysis. Regression remains one of political science’s most widely used tools because it provides a tractable, flexible, and interpretable framework for connecting theory with data. While traditional regression analyses typically rely on survey responses or behavioral measures like voting records, incorporating political objects as variables opens new opportunities to study how visual elements in political imagery shape attitudes, behaviors, and outcomes.

When the subjects of interest are images, constructing variables that capture politically meaningful concepts becomes far more challenging, as scholars cannot directly query images about their latent content. Human annotation is one solution, but it is costly—requiring either experts or extensive training for non-experts—and even when labels are assigned, it is often unclear which specific visual features annotators relied on. This combination of expense and opacity makes it difficult to ensure the consistency and validity of annotated variables.

Off-the-shelf object detection tools alleviate some of these issues by reliably identifying semantically meaningful objects such as “person,” “car,” or “building.” Yet these categories are typically too broad to address political science questions. Protest researchers, for instance, are not simply interested in whether an image depicts people, but whether those individuals are protesters, or police officers—subcategories with distinct theoretical implications. Likewise, studying protest violence

requires distinguishing between state and protester aggression, a task beyond the capabilities of generic detection models. Because such tools stop at coarse categories, they cannot explain the visual signals driving human perceptions of political meaning.

My approach addresses this gap by clustering and labeling visual segments into politically salient categories that are both semantically coherent and theoretically relevant. This process generates interpretable variables that allow political scientists to map concrete visual features—such as state presence, protester identity, or signs of violence—to abstract concepts that matter for theory building and hypothesis testing. In this application, I illustrate how these political object variables can be incorporated into regression analyses to study media bias and its potential effects.

4.2 Analysis

Media bias has long been linked to polarization, with two dominant strands of research. The first examines the existence of bias, typically focusing on textual data because scalable methods for analyzing visuals have been lacking. The second investigates the effects of bias through experiments in which participants are exposed to liberal, conservative, or neutral content and their attitudes toward political issues are measured.

Both strands face important limitations. Text-focused studies overlook the visual dimension of media coverage, even though images often carry strong political signals. Experimental studies, meanwhile, rarely evaluate whether the visual stimuli they use are representative, nor do they systematically disentangle which components of the stimuli drive the observed effects. These gaps limit our understanding of both the scope and the mechanisms of media bias.

The proposed framework addresses both of these shortcomings. In the first analysis, I test whether a conservative outlet portrays BLM protests as involving greater police presence than a liberal outlet. Using my customized political object detection tool, I construct two dependent variables: (a) a binary indicator of police presence and (b) a continuous measure of the police-to-protester ratio. I then regress each outcome on the ideological alignment between the protesters and the media outlet. The results show that the conservative outlet systematically depicts BLM

protests with heavier police presence. Importantly, constructing such dependent variables would not be possible with off-the-shelf object detection tools, which cannot separately identify protesters and police officers, and would be prohibitively costly to generate through human annotation.

The second analysis shifts from documenting the existence of bias to examining one of its potential effects: whether the portrayal of police presence influences perceptions of protest violence. Here, I again use measures of police presence and the police-to-protester ratio derived from my detection tool and test whether these visual cues predict perceived violence, drawing on predictions from the protest violence model developed by (Won, Steinert-Threlkeld, and Joo 2017). The findings indicate that images depicting police and fire heighten perceived violence, whereas images featuring defensive gear such as shields reduce it. Together, these analyses demonstrate how political object detection can reveal both the presence of bias in visual media and its potential consequences for public opinion.

4.3 Data

I collect front-page images from Wall Street Journal (WSJ) and New York Times (NYT) articles covering BLM protests that followed the death of George Floyd in late May 2020, with the dataset spanning through the end of June 2020. I focus on BLM protests because the movement has several characteristics that make it particularly well suited for testing. Despite belonging to the same movement, the protests took many different forms across time and place, offering photojournalists a wide variety of images to choose from. This abundance of visual material creates an ideal setting to study media bias, since image selection is less constrained by limited availability. By contrast, if only a handful of images existed, even the most ideological journalist might struggle to consistently highlight their preferred protesters in a favorable light.

I select the WSJ and the NYT for practical reasons: both outlets maintain publicly accessible news archives, and they are comparable in readership size and ideological orientation. Among the 730 images collected, 530 are from the NYT and 200 from the WSJ. Whether this imbalance reflects differences in newsroom capacity, editorial choices, or some combination of the two lies

beyond the scope of this paper.

4.3.1 Variables

Ideological Agreement is a binary variable indicating whether the political orientation of the source media outlet aligns with that of the protesters. Since all images in the dataset are drawn from BLM protests, the variable takes the value 1 if the image comes from the New York Times (NYT) and 0 if it comes from the Wall Street Journal (WSJ).

Perceived Violence is a continuous variable ranging from 0 to 1, constructed using the protest violence prediction model developed by (Won, Steinert-Threlkeld, and Joo 2017). Higher values indicate greater levels of perceived violence in the image. For the purposes of this analysis, the model’s predictions are treated as ground truth, though, as Egami et al. (2025) caution, this assumption may lead to biased estimates even when the underlying model is highly accurate.

Police Presence is a binary variable indicating whether at least one police officer appears in the image. The variable takes the value 1 if an officer is present and 0 otherwise. The number of officers is estimated from the presence of police-associated headwear, specifically police hats, helmets, and white bike helmets. This provides a reasonably precise estimate, since officers typically signal their authority through uniformed attire, and only rarely appear without some form of headwear.

Police-to-Protesters Ratio is a continuous variable ranging from 0 to 1, calculated by dividing the estimated number of police officers by the number of protesters. The number of protesters is obtained by subtracting the number of police officers from the total number of detected faces. This provides a reasonably precise estimate, as contemporary face-detection tools perform at a level comparable to human detection.

Fire is measured in two forms: a binary indicator and a continuous variable. The binary indicator takes the value 1 if any visible fire is detected in the image and 0 otherwise. The continuous variable, *Fire Size*, captures the proportion of the image area occupied by fire. These variables serve as direct proxies for violent protest dynamics, as fire is a highly visible signal of escalation.

Shields are likewise measured in both binary and continuous forms. The binary variable takes

the value 1 if at least one shield is visible and 0 otherwise, while the continuous variable, *Shield Count*, records the number of shields detected. Shields are classified as defensive gear, and their presence may reduce rather than increase perceptions of violence, depending on the protest context.

Batons are operationalized similarly. The binary variable takes the value 1 if any baton is detected and 0 otherwise, while the continuous measure, *Baton Count*, records the number of batons. Batons are categorized as offensive gear, and thus are expected to be positively associated with perceptions of violence.

Control variables include *Maximum Face Size*, *Protester Count*, and *Black-to-Non-Black Ratio*. *Maximum Face Size* is a continuous variable calculated by dividing the area of the largest detected face by the overall image size. *Protester Count* is estimated by subtracting the number of identified police officers from the total number of detected faces. *Black-to-Non-Black Ratio* is computed by dividing the number of predicted Black faces by the number of predicted non-Black faces, with race classifications generated using the Python DeepFace library. I include these three controls in the second analysis as they are likely to influence either or both *Police Presence* (the key independent variable) and *Perceived Violence* (the outcome). However, they are not included in the first analysis, where the key independent variable is *Ideological Agreement* as the three variables are post-treatment variables that occur as a result of ideological agreement (or lack thereof).

4.4 Results

4.4.1 Analysis 1: Existence of Bias

Table 7 reports four regression models. The first model uses all 731 images in the dataset, while the remaining three models restrict the sample to the 28 images featuring at least one police officer. Across all specifications, the independent variable of interest is *Ideological Agreement*, while the dependent variable varies: *Police Presence* (Model 1), *Police-to-Protester Ratio* (Model 2), *Baton Presence* (Model 3), and *Shield Presence* (Model 4). The results indicate that ideological agreement between protesters and the media outlet has a statistically significant negative effect on the likelihood that an image depicts any police officer (Model 1). By contrast, conditional on police

Table 7: Analysis 1 Regression Results

	Police Presence	Police-to-Protesters Ratio	Baton Presence	Shield Presence
Ideological Agreement	−1.339*** (0.391)	−1.480 (0.899)	0.788 (0.790)	−0.310 (0.858)
Num.Obs.	731	28	28	28
R2		0.094		
R2 Adj.		0.060		
AIC	229.9	131.3	40.5	37.4
BIC	239.1	135.3	43.2	40.0
Log.Lik.	−112.974	−62.666	−18.255	−16.685
F	11.707	2.712	0.996	0.131
RMSE	0.19	2.27	0.48	0.45

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

being present in the image, ideological agreement does not significantly predict the relative number of police to protesters, nor the probability that batons or shields appear (Models 2–4).

4.4.2 Analysis 2: Effect of Bias

Table 8 reports five regression models where the dependent variable across all specifications is Perceived Violence. Model 3 restricts the sample to the 28 images featuring at least one police officer, while the other four models use all 731 images in the dataset. The key independent variables vary by model: Police Presence (Model 1), Police-to-Protester Ratio (Models 2–3), and combinations of Baton, Shield, and Fire variables (Models 4–5).

Models 1 and 2 show that both Police Presence and Police-to-Protester Ratio have statistically significant positive effects on Perceived Violence. However, the adjusted R-squared values indicate that Police Presence explains more variation in perceived violence than Police-to-Protester Ratio. In Model 3, when restricting to images that already contain police officers, the Police-to-Protester Ratio no longer has a significant effect, suggesting that the simple presence of police officers, rather than their relative proportion, is what matters for perceptions of violence.

Models 4 and 5 introduce additional protest features. In Model 4, Shield Presence has a statis-

tically significant negative effect on Perceived Violence, while Fire Presence has a strong positive effect. Baton Presence, however, shows no significant relationship. The inclusion of these features substantially increases the explanatory power of the model, doubling and tripling the adjusted R-squared compared to Model 1 and 2, respectively. This result aligns with intuition: shields function as defensive gear, dampening perceptions of violence, whereas fire is strongly associated with heightened violence.

Model 5 replaces the binary indicators of Model 4 with continuous measures: Shield Count, Baton Count, and Fire Size. The results again show that Shield Count reduces, and Fire Size increases, perceived violence, both with high statistical significance. Baton Count remains insignificant. Substantively, these findings suggest that while the presence of shields and fire shapes perceived violence, the quantity of shields or the size of fire does not further alter perceptions beyond the binary presence. Moreover, the adjusted R-squared values of Models 4 and 5 are nearly identical, indicating that shifting from binary to continuous measures does not improve model fit in this case.

Analyses 1 and 2 show that simple binary indicators for police, shield, baton, and fire presence explain perceived violence in protest images as effectively as more granular continuous measures of the same features. This suggests that, in many cases, scholars may only need binary indicators, which can be obtained by training straightforward image classifiers for each feature. Classifiers are easier to build and deploy than object detection models. However, object detection remains essential because it produces bounding boxes that allow us to extract continuous measures—such as the number of shields, the ratio of police to protesters, or the size of fires—which cannot be derived from classifiers alone. While in this study binary indicators suffice, in other contexts continuous measures may capture important variation that binary indicators miss. Crucially, researchers cannot know in advance whether binary or continuous features better explain an outcome. Thus, developing object detection tools that yield granular continuous measures is always preferable, since these measures can easily be reduced to binary indicators if needed, but the reverse is not possible.

Table 8: Analysis 2 Regression Results

	(1)	(2)	(3)	(4)	(5)
Max Face Size	0.001 (0.013)	−0.002 (0.013)	−0.150+ (0.077)	−0.012 (0.013)	−0.013 (0.013)
Crowd Count	0.006*** (0.001)	0.006*** (0.001)	−0.001 (0.006)	0.006*** (0.001)	0.006*** (0.001)
Black to Nonblack Ratio	−0.028+ (0.016)	−0.036* (0.016)	−0.126 (0.183)	−0.029+ (0.015)	−0.029+ (0.015)
Police Binary	0.165*** (0.026)			0.157*** (0.025)	0.160*** (0.025)
Police Ratio		0.030** (0.010)	−0.003 (0.011)		
Shield Binary				−0.064*** (0.011)	
Baton Binary				0.008 (0.015)	
Fire Binary				0.194*** (0.025)	
Shield					−0.036*** (0.008)
Baton					0.002 (0.009)
Fire					0.158*** (0.020)
Num.Obs.	731	731	28	731	731
R2	0.111	0.073	0.159	0.208	0.199
R2 Adj.	0.106	0.068	0.013	0.201	0.191
AIC	−868.5	−838.0	−29.9	−947.0	−938.5
BIC	−840.9	−810.4	−21.9	−905.7	−897.1
Log.Lik.	440.233	424.996	20.946	482.525	478.232
F	22.700	14.363	1.087	27.173	25.650
RMSE	0.13	0.14	0.11	0.13	0.13

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

5 Discussion

5.1 Contributions

This paper makes several contributions to the study of computational social science and political communication. First, I introduce a novel domain knowledge-guided cluster-then-label framework for detecting politically salient objects in images. By combining open-vocabulary object detection, segmentation, unsupervised clustering, and domain knowledge-guided labeling, the framework avoids the cost of full-scale manual annotation while producing semantically coherent and theoretically meaningful object categories. This addresses a major gap between off-the-shelf computer vision tools—often limited to generic objects like “person” or “car”—and the nuanced needs of political scientists interested in symbolic and interpretive dimensions of visual politics.

Second, I develop a quantitative evaluation strategy to assess the accuracy and relevance of the customized detection tool. This evaluation leverages domain knowledge to create both qualitative and statistical measures—such as visual coherence and modified F1 scores—that capture not only technical performance but also theoretical validity. This contribution ensures that the detection framework can be rigorously validated.

Third, the entire pipeline is designed to be transparent and financially accessible, relying exclusively on open-source models. Because it does not require expensive proprietary software, commercial APIs, or specialized hardware, the framework can be adopted by researchers at institutions with limited computational resources or funding. This emphasis on transparency and scalability ensures that cutting-edge computational methods for political image analysis remain widely available across the discipline, rather than restricted to a small number of well-funded labs.

Finally, I extend the framework through two substantive applications. The first introduces *Objscores*, a systematic and scalable approach to quantifying the ideological valence of visual political objects. These scores enable new lines of research on symbolic politics, framing, and visual persuasion, including longitudinal analyses of how symbols evolve over time and experiments isolating the ideological cues embedded in images. The second application integrates detected politi-

cal objects into regression analysis, showing how visual political cues can be linked to theoretical constructs such as media bias and perceived protest violence. This demonstrates the framework’s potential to bridge computational image analysis with widely used statistical methods in political science.

5.2 Limitations

Despite its contributions, the proposed framework faces a key limitation related to scalability of manual evaluation. Although the domain knowledge-guided cluster labeling approach reduces annotation costs, it still requires expert judgment to validate and label algorithm-generated clusters. Scaling this process to accommodate a larger number of object categories or diverse image types would demand substantial time and labor, potentially limiting its applicability in large-scale studies.

The first application also faces its own limitation. The estimated *Objectscores* lack confidence intervals, making it impossible to determine whether one object is statistically more liberal or conservative than another. Although constructing such intervals is beyond the scope of this paper, future work could incorporate resampling-based methods, such as bootstrapping, to quantify uncertainty in the estimated scores.

Finally, both the first and second applications share a common limitation: they assume that the customized political object detection model produces error-free predictions. However, as shown by Egami, Hinck, Stewart, and Wei (2025), treating machine learning predictions as ground truth can lead to invalid estimates and misleading confidence intervals, regardless of the model’s accuracy. Incorporating their recommended methods for uncertainty quantification would strengthen the validity of future analyses.

6 Conclusion

This paper introduces a domain knowledge-guided cluster-then-label framework for detecting politically meaningful objects in images, addressing critical gaps in flexibility, cost, and applicability

faced by existing computer vision tools. By combining open-vocabulary detection, segmentation, unsupervised learning, and lightweight classification models, the framework moves beyond generic object detection to capture theoretically relevant visual features in real-world political imagery. Importantly, it provides a scalable and transparent approach that remains accessible to researchers without extensive funding or specialized computational resources.

The two applications demonstrate how this framework expands the methodological toolkit for political science. *Objectscores* enable systematic, and scalable measurement of the ideological valence of political objects, opening new avenues for studying symbolic politics, framing, and visual persuasion. The second application incorporates detected political objects as variables in regression models, linking politically visual features in media coverage to theories of media bias and perceptions of protest violence. Together, these applications illustrate how the proposed framework integrates computational image analysis with established empirical strategies, creating new opportunities for multimodal research on political communication.

7 Appendix

Query	Image Count	Segment Count
2017 Women’s March	270	1571
January 6 Capitol Attack	223	2259
Police protester confrontation	209	1095

Table 9: Image and segment counts by query.

Algorithm 1 Labeling strategy for clustered samples

Require: Cluster assignments and originating query metadata

Ensure: Label for each sample

```
1: for all samples do
2:   if Cluster  $\in \{-1, 2\}$  then
3:     Label  $\leftarrow$  "Unknown"
4:   else if Cluster = 1 then
5:     Label  $\leftarrow$  "Police"
6:   else if Cluster = 0 then
7:     if Query = '2017 Women's March' then
8:       Label  $\leftarrow$  "Pussy"
9:     else if Query = 'January 6 Capitol Attack' then
10:      Label  $\leftarrow$  "MAGA"
11:   end if
12: end if
13: end for
```

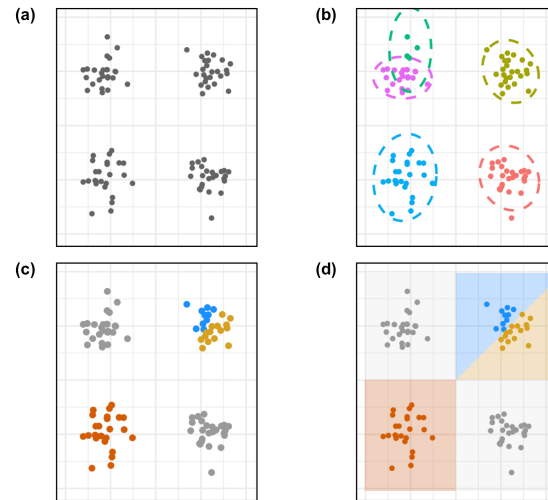


Figure 8: A visualization of domain knowledge-driven cluster-then-label approach. (a) Unlabeled raw segments; (b) Clustering results; (c) Domain knowledge-driven labeling; (d) Predictions

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